

IRIS Flower Classification using Logistic Regression and Random Forest

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Abstract

This project is part of the Foundational Summer Internship Program 2026 at IDEAS (Institute of Data Engineering, Analytics and Science Foundation) - TIH, ISI Kolkata. It tackles the classic machine learning problem of classifying Iris flowers into three species: setosa, versicolor, and virginica. The dataset used is the well-known Iris dataset, containing 150 samples with four features each — sepal length, sepal width, petal length, and petal width. The project is structured as a guided, question-answer style notebook, walking through each step of the ML pipeline. It begins with loading and exploring the dataset, then organizes it into a Pandas DataFrame for easy manipulation. The data is then split into training and testing sets in an 80/20 ratio using scikit-learn's `train_test_split`. Two classification models are trained and compared: Logistic Regression as a linear baseline and Random Forest as an ensemble method. The Logistic Regression model is evaluated using accuracy score, while the Random Forest model (with 100 estimators) is assessed using a full classification report. The classification report provides precision, recall, and F1-score for each of the three flower classes. The side-by-side comparison of Logistic Regression and Random Forest makes it a particularly instructive exercise in understanding model trade-offs.

Introduction

This project focuses on classifying Iris flowers into three species using machine learning, a problem that has remained a benchmark in the data science community since Ronald Fisher introduced the dataset in 1936. Its enduring relevance lies in how well it illustrates core classification concepts in a clean, interpretable setting. The project employs Python as its primary language, leveraging key libraries such as Pandas for data manipulation and scikit-learn for model building and evaluation. Two algorithms are explored — Logistic Regression, a foundational linear classifier, and Random Forest, a powerful ensemble method — offering a natural basis for comparing simple versus complex models. The primary goal is to strengthen interns' understanding of the supervised learning pipeline, from raw data to model evaluation, using tools that are standard across the industry.

During the first two weeks of our summer internship here at IDEAS-TIH, classes were held to give us a first hand understanding of core Python concepts like learning to write code from scratch to discovering extensive libraries like Pandas and NumPy. Also we later explored the concepts of Machine Learning, various models being trained and tested, their significance etc. and had interactive sessions being held on the topics of LLMs and Communication Skills with agentic AI.

Project Objectives

Introduce the ML Pipeline: To familiarize ourselves with the complete supervised machine learning workflow — from data loading and exploration to model training and performance evaluation — using a clean, well-structured dataset.

Build Hands-on Proficiency with Tools: To develop practical skills with industry-standard Python libraries such as Pandas for data manipulation and scikit-learn for building, training, and evaluating classification models.

Understand and Compare Classification Algorithms: To provide a side-by-side learning experience of two fundamentally different classifiers — Logistic Regression as a linear baseline and Random Forest as an ensemble approach — helping ourselves appreciate the trade-offs between model simplicity and complexity.

Develop Model Evaluation Skills: To learn how to assess model performance meaningfully using metrics like accuracy score, precision, recall, and F1-score, and to understand what these metrics reveal about a model's real-world reliability.

Bridge Theory and Practice: To translate classroom knowledge of machine learning concepts into a working, reproducible project, preparing ourselves for the kind of structured problem-solving expected in professional data science and analytics roles.

Methodology

The project follows a structured, step-by-step supervised machine learning methodology as laid out in the Jupyter notebook. It begins with importing necessary libraries, followed by loading and describing the Iris dataset using scikit-learn's built-in `load_iris()` function. The dataset is then converted into a Pandas DataFrame for easier manipulation and exploration. A target column is added and its class distribution is verified before separating the data into feature matrix X and target vector Y. The notebook then proceeds to split the data, train two classification models sequentially, and evaluate each using appropriate metrics - forming a clean, reproducible analytical pipeline from raw data to final model assessment.

Two classification models were selected for this project — Logistic Regression and Random Forest — chosen deliberately to represent contrasting approaches to the same problem.

Logistic Regression was selected as the baseline model due to its simplicity, interpretability, and suitability for linearly separable data. Since the Iris dataset contains classes that are largely separable by linear boundaries (especially setosa), it was a natural first choice. Logistic Regression was validated using accuracy score, which is a reliable metric here given the perfectly balanced class distribution of 50 samples per class, eliminating any risk of the metric being misleading due to class imbalance.

Random Forest was selected as the more advanced model to demonstrate the power of ensemble learning. It builds multiple decision trees on random subsets of the data and aggregates their predictions, making it robust, non-linear, and resistant to overfitting — a strong contrast to Logistic Regression. Random Forest was validated using a classification report, which provides a more granular breakdown of performance — reporting precision, recall, and F1-score for each of the three flower classes (setosa, versicolor, virginica) individually.

Both models were trained exclusively on the training set and evaluated on the held-out test set, ensuring that the validation reflects true generalization performance on unseen data rather than in-sample fit.

The use of `random_state = 42` in both the train-test split and the Random Forest instantiation ensures full reproducibility of results across different runs and environments.

Together, the two models illustrate the trade-off between simplicity and complexity, and between linear and non-linear decision boundaries.

The dataset of 150 samples was split into training and testing sets using scikit-learn's `train_test_split` function with the following parameters:

Test size: 20% (30 samples reserved for testing)

Training size: 80% (120 samples used for training)

Random state: 42, ensuring the split is fixed and reproducible across runs

The same split was used for both models, ensuring a consistent and fair basis for comparison, with no data leakage between training and testing phases.

The following is the github repository link for the solved jupyter notebook:

https://github.com/Ritobroto20061612/IRISFlowerSummerProject/blob/31fbc5c18039b490261fed000998ce6aa9d4f8a7/Copy_of_01_IRIS_Flower_Classification_using_Logistic_Regression_and_Random_Forest_Summer_2026.ipynb

Data Analysis and Results

Descriptive Analysis

The Iris dataset consists of 150 samples evenly distributed across three flower species — setosa, versicolor, and virginica — with exactly 50 samples per class, making it a perfectly balanced dataset. Each sample is described by four numerical features: sepal length, sepal width, petal length, and petal width, all measured in centimetres. The dataset was loaded directly from scikit-learn's built-in library and converted into a Pandas DataFrame for exploration. Upon checking the class distribution using `value_counts()`, the equal split of 50 samples per class was confirmed, indicating no class imbalance concerns. The data was then separated into a feature matrix `X` of shape (150, 4) and a target vector `y` of shape (150,), before being split into a training set of 120 samples and a testing set of 30 samples using an 80/20 ratio with `random_state=42`.

Inferential Analysis

Both models were trained on the 120-sample training set and evaluated on the 30-sample test set. Logistic Regression, evaluated using accuracy score, achieved a perfect score of 1.0, correctly classifying all 30 test samples. Random Forest, evaluated using a full classification report, similarly achieved perfect precision, recall, and F1-score of 1.00 across all three classes, with a macro and weighted average of 1.00. These results suggest that the Iris dataset is sufficiently well-structured and linearly separable enough for even a simple baseline model like Logistic Regression to achieve perfect classification. The fact that Random Forest matched this performance confirms that the dataset does not present any meaningful complexity that would require an ensemble approach to resolve. Overall, both models generalise perfectly to unseen data in this context, though it should be noted that such results are characteristic of the Iris dataset's clean nature rather than a reflection of either model's superiority over the other.

Model Comparison

Both Logistic Regression and Random Forest were trained and evaluated on the same Iris dataset using an identical 80/20 train-test split, ensuring a fair and consistent basis for comparison.

Logistic Regression performed remarkably well, achieving a perfect accuracy score of 1.0 (100%) on the 30-sample test set. This result is not entirely surprising given that the Iris dataset, particularly the setosa class, is largely linearly separable. The model converged quickly due to its low computational complexity, and its predictions on all 30 test samples were correct. Its strength in this context lies in the simplicity and clarity of its decision boundary, making it easy

to understand and interpret. However, it is worth noting that on more complex or overlapping datasets, Logistic Regression would likely struggle where a non-linear model would not.

Random Forest, with 100 decision trees and a fixed random state of 42, also achieved perfect scores across all evaluation metrics. The classification report confirmed a precision, recall, and F1-score of 1.00 for each of the three classes — setosa, versicolor, and virginica — with a macro and weighted average of 1.00 across all 30 test samples. This indicates that the ensemble approach correctly identified every flower in the test set without a single misclassification. The model's ability to handle non-linear boundaries and feature interactions makes it inherently more powerful, though that additional complexity was not strictly necessary for a dataset as clean and well-structured as Iris.

Overall, both models delivered identical predictive accuracy on this dataset, which reflects the fact that the Iris dataset is well-suited for classification and does not pose significant challenges to either approach. The key differentiator between the two is not performance on this particular task, but rather their underlying mechanisms and scalability — Logistic Regression is faster, more interpretable, and sufficient for linearly separable problems, while Random Forest is more robust, flexible, and better equipped to handle real-world datasets where noise, non-linearity, and feature interactions are more prevalent. For the purpose of this internship project, the comparison effectively illustrates that model selection should be driven not just by accuracy, but by the nature of the data and the complexity of the problem at hand.

Conclusion

The primary conclusion drawn from this project is that both Logistic Regression and Random Forest are highly effective classifiers for the Iris flower classification problem, with both models achieving a perfect accuracy of 100% on the held-out test set. This outcome validates the choice of models and confirms that the Iris dataset, with its balanced class distribution of 50 samples per class and four well-discriminating features, is well-suited for supervised classification tasks.

The perfect accuracy of Logistic Regression — a linear, parametric model with default settings — strongly suggests that the three Iris species are largely linearly separable in the four-dimensional feature space. This is further supported by the fact that no feature engineering, hyperparameter tuning, or data preprocessing was required to achieve this result, indicating that the raw features themselves carry sufficient discriminatory power.

Random Forest's perfect classification report, with precision, recall, and F1-score of 1.00 across all three classes - setosa, versicolor, and virginica, corroborates this finding from a non-linear perspective. The fact that a more complex ensemble method did not outperform the simpler baseline further reinforces the conclusion that the dataset does not harbour any non-linear complexity that would necessitate a more sophisticated approach.

It can also be concluded that the 80/20 train-test split with random_state=42 provided a reliable and reproducible evaluation framework, ensuring that the reported results are consistent and not a product of a favourable random partition of the data.

APPENDICES

1. Solved Jupyter Notebook:-

https://github.com/Ritobroto20061612/IRISFlowerSummerProject/blob/31fbc5c18039b490261fed000998ce6aa9d4f8a7/Copy_of_01_IRIS_Flower_Classification_using_Logistic_Regression_and_Random_Forest_Summer_2026.ipynb

2. Copy of this report:-